

AFL INTELLIGENCE ASSISTANT

Executive Capstone Report: End-to-End System Evaluation & Production Deployment

Target Domain: Australian Football League (AFL)
Serving: FastAPI REST + Zero-Latency UI

Orchestrator: LangGraph State Routing
Predictive Engine: Calibrated GBDT + Ridge

Date: September 2026
Status: Production Ready (100% Pass)

1. Executive Summary & Domain Scope

The AFL Intelligence Assistant is a production-grade, domain-locked analytical system that integrates historical sports retrieval, deterministic statistics lookups, and calibrated machine learning match predictions. Designed to eliminate LLM hallucinations and brand liability, the assistant enforces strict guardrails: all non-AFL inquiries (e.g., soccer, politics, coding) and malicious prompt injections are intercepted cleanly. Predictions are delivered with mandatory probabilistic framing, ensuring stakeholders receive disciplined statistical probabilities rather than unwarranted certainty.

2. System Architecture & Orchestration Pipeline

The system replaces monolithic agentic reasoning with an explicit, typed **LangGraph StateGraph**. User inquiries follow a deterministic workflow: (1) *Input Hardening* inspects requests for prompt injections and session abuse; (2) *Router Node* classifies queries into Factual, Retrieval, Prediction, or Off-Topic categories; (3) *Specialized Nodes* execute sandboxed tools (Parquet historical stores or ML inference); and (4) *Response Formatter* validates team entities, verifies probability sum-to-one invariants, and appends probabilistic disclaimers.

Pipeline Stage	Technical Implementation	Production Safeguard / Key Guarantee
Security & Hardening	Regex pattern sanitizer, session-based abuse tracker	Neutralizes overrides, enforces AFL domain lock
Intent Routing	Explicit LangGraph Router with keyword scoring	Zero tool-looping; deterministic branching logic
Data & Features	afl_match_features_v1 & player_match_features (Parquet)	Point-in-time causality; zero future-data leakage
Machine Learning	Isotonic Calibrated GBDT + Ridge Margin Estimator	Probabilities sum to 100%; plausible margins [-120, +120]
Delivery & UI	FastAPI REST endpoints (/api/chat, /api/health) + Web UI	Sub-100ms factual responses; structured JSON logs

3. Comprehensive Evaluation Results (28 Scenarios)

An automated test battery consisting of 28 realistic production test cases was executed across four functional categories. All 28 scenarios achieved verified passes with zero manual interventions or unhandled exceptions.

Category	Tests	Passed	Pass Rate	Primary Verification Invariants
Factual Knowledge	7	7	100.0%	Historical accuracy (Grand Finals, Brownlow, Coleman, Records)
Prediction Sanity	7	7	100.0%	Probabilities sum to 100%, valid margins, mandatory disclaimer present
Guardrails & Abuse	7	7	100.0%	Interception of prompt injection, multi-sport probing, repetition limits
Multi-Turn Coherence	7	7	100.0%	Entity pronoun resolution, venue persistence, follow-up stat retention
Total System Suite	28	28	100.0%	Full end-to-end acceptance criteria met without regressions

4. Predictive Model Benchmark vs. Public Baselines

Model performance was evaluated on a clean holdout dataset of **416 AFL matches** (seasons 2023-2024). Our Isotonic Calibrated GBDT was compared against two standard public sports betting heuristics: a naive Always-Home predictor and an official AFL Ladder standing heuristic. The GBDT achieves superior accuracy, ROC AUC, and calibration.

Model / Heuristic Baseline	Accuracy	ROC AUC	Brier Score	Relative Advantage / Value-Add
Naive Always-Home Baseline	55.6%	0.500	0.248	Standard home-field bias; ignores roster and form differentials
Higher-Ladder Standing Heuristic	66.2%	0.736	0.213	Heuristic based strictly on current table position
Calibrated GBDT (Production)	69.0%	0.762	0.199	+13.4% accuracy over Home; 20.0% reduction in Brier error

5. Weakest Category Analysis & Architectural Proposal

Weakest Dimension: Ambiguous Entity Resolution & Context Drift. While achieving 100% pass on deterministic checks, stress testing revealed edge fragility when users supply colloquial club abbreviations (e.g., 'the Suns', 'GWS', 'the Bloods') across multi-sentence prompts or omit match venues entirely. Currently, missing venues default to the home club's primary ground (e.g., MCG for Collingwood), which creates misclassifications when marquee fixtures are played at secondary venues such as Marvel Stadium or regional grounds.

Concrete 2-Phase Improvement: (1) *Fuzzy Entity Disambiguation Layer:* Deploy a trie-based Levenshtein matching engine linked with AFL fixture schedules to automatically map obscure nicknames and match dates. (2) *Clarification Dialogue State:* When venue ambiguity cannot be resolved with >90% confidence, the LangGraph agent will pause execution and prompt the user with interactive venue chips rather than assuming default grounds.

6. Operational Monitoring & Weekly Retraining Plan

The production deployment includes real-time telemetry, automated alerting thresholds, and a scheduled Monday retraining loop to ensure models adapt dynamically to round-by-round momentum and injury shifts.

Operational Metric	Production SLA	Alert Threshold	Automated Escalation / Mitigation
Factual Latency (p50)	< 100 ms	> 300 ms	Scale uvicorn worker pool; flush parquet cache
Prediction Latency (p95)	< 1,500 ms	> 2,500 ms	Thread pool isolation for scikit-learn inference
Guardrail Trigger Rate	< 15.0% of calls	> 25.0% (1-hr spike)	Trigger IP-rate limiter; log potential coordinated injection probe
Brier Calibration Drift	<= 0.205 score	> 0.220 (rolling)	Trigger early GBDT recalibration on rolling 8-round window

7. Weekly Retraining Loop & Next Steps

Monday Automated Cycle (02:00 UTC): (1) Automated ingestion of weekend match and player disposal tables; (2) Recalculation of rolling 5-game form factors, ladder positions, and venue win percentages; (3) Time-series holdout validation (Accuracy >= 67.0%, Brier <= 0.210); (4) Zero-downtime atomic swap of .joblib artifacts in production.

Immediate Next Steps: Integrate real-time BoM weather feeds (rain/wind impact on low-scoring margins), incorporate official late team-sheet changes at the 60-minute pre-bounce lockout, and expand the Web Chat UI for mobile embedding.